

INVENTORY MANAGER

A SIMPLE INVENTORY MANAGEMENT SYSTEM WITH DATABASE INTEGRATION

Nikhil Ingole

PROJECT OVERVIEW

1. Objective:

- Build an inventory management system to handle adding, updating, deleting, and searching inventory items.
- Integrate MySQL for database management.

2. Features:

- View Inventory Items
- Add, Update, and Delete Items
- Search Items by Name
- Store and retrieve data from a MySQL database

DATABASE SCHEMA

- Database Name: inventory_db
- Table:
 - items
 - Columns: id, name, quantity, price
 - Purpose: Stores details of the inventory items.

	id	name	quantity	price
▶	1	Pen	50	10
	2	Pencil	50	5
	3	Book	100	50
✱	NULL	NULL	NULL	NULL

SYSTEM ARCHITECTURE

- Components:
- Frontend: User Input via the command line interface (CLI).
- Backend: Java Application (JDBC) communicates with the MySQL database.
- Database: MySQL to store inventory data.
- Process Flow:
 1. User interacts with the app through the CLI.
 2. App queries and updates the database as per user input.
 3. Data is shown on the console.

MAIN FEATURES

- **View Items:** Displays a list of all items stored in the inventory.
- **Add Item:** Adds new items to the inventory with name, quantity, and price.
- **Update Item:** Updates the quantity and price of an existing item.
- **Delete Item:** Deletes an item from the inventory by ID.
- **Search Items:** Search for inventory items by their name.

METHOD OVERVIEW

Method	Purpose	Main SQL Query
viewItems	Display all items	SELECT * FROM items
addItem	Add new item	INSERT INTO items (name, quantity, price)
updateItem	Update existing item	UPDATE items SET quantity = ?, price = ? WHERE id = ?
deleteItem	Delete item by ID	DELETE FROM items WHERE id = ?
searchItems	Search item by name	SELECT * FROM items WHERE LOWER(name) LIKE ?

DATABASE

1. CREATE DATABASE inventory_db;
2. USE inventory_db;
3. CREATE TABLE items (- 4. id INT PRIMARY KEY AUTO_INCREMENT,
- 5. name VARCHAR(100) NOT NULL,
- 6. quantity INT NOT NULL,
- 7. price DOUBLE NOT NULL
- 8.);

```
4. Delete Item
5. Search Items by Name
6. Exit
Choose an option: 1
ID      Name      Quantity      Price
1       Book       250          200.00
2       Pencil     60           5.00
3       Eraser     50           5.00
```

```
Inventory Menu:
1. View Items
2. Add Item
3. Update Item
4. Delete Item
5. Search Items by Name
6. Exit
Choose an option: 3
Enter item ID to update: 1
Enter new quantity: 300
Enter new price: 199.9
Item updated.
```

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Inventory Menu:
1. View Items
2. Add Item
3. Update Item
4. Delete Item
5. Search Items by Name
6. Exit
Choose an option: 5
Enter name to search for: Pencil
ID      Name      Quantity      Price
2       Pencil     60           5.00
```

```
Inventory Menu:
1. View Items
2. Add Item
3. Update Item
4. Delete Item
5. Search Items by Name
6. Exit
Choose an option: 2
Enter name: Pen
Enter quantity: 100
Enter price: 10
Item added with ID: 4
```

```
2. Add Item
3. Update Item
4. Delete Item
5. Search Items by Name
6. Exit
Choose an option: 1
ID      Name      Quantity      Price
1       Book       300          199.90
2       Pencil     60           5.00
3       Eraser     50           5.00
4       Pen        100          10.00
```

```
Inventory Menu:
1. View Items
2. Add Item
3. Update Item
4. Delete Item
5. Search Items by Name
6. Exit
Choose an option: 4
Enter item ID to delete: 4
Item deleted.
```

```
Inventory Menu:
1. View Items
2. Add Item
3. Update Item
4. Delete Item
5. Search Items by Name
6. Exit
Choose an option: 1
ID      Name      Quantity      Price
1       Book       300          199.90
2       Pencil     60           5.00
3       Eraser     50           5.00
```

```
Inventory Menu:
1. View Items
2. Add Item
3. Update Item
4. Delete Item
5. Search Items by Name
6. Exit
Choose an option: 6
Goodbye!
PS C:\Users\aditya\Documents\Programs\Inventory Management>
```


CONCLUSION AND FUTURE WORK

- **Conclusion:**
- Successfully implemented a basic inventory management system with Java and MySQL.
- **Future Enhancements:**
 1. Implementing a GUI for easier interaction.
 2. Adding categories and suppliers for better organization.
 3. Generating reports for inventory statistics.



THANK YOU